

Use-Case Flow

Main Success Scenario:

1. **Select Option:** Faculty Researcher selects "Log Expense" from the grant management portal.
2. **Input Details:** System prompts for Expense Item Name, Category, Vendor Name, Transaction Date, Amount, and Supporting Invoice PDF.
3. **Submit Data:** Faculty Researcher fills in required fields, attaches invoice, and clicks "Submit Expense".
4. **Validate Input:** System verifies required fields and attachment format validity.
5. **Check Balance:** System queries current available balance for specified grant.
6. **Verify Funds:** System confirms requested amount does not exceed remaining budgetary balance.
7. **Deduct & Update:** System deducts amount from allocated budget and updates remaining balance in real time.
8. **Audit Logging:** System writes transaction details to immutable audit ledger.
9. **Confirmation:** System displays success message with unique Expense Reference ID.
10. **End:** Use case ends successfully.

Alternate Flows:

- **6a. Requested Expense Exceeds Remaining Balance (<<extend>> UC-07: Flag Overdraft Warning)**
 - **6a1.** System detects amount exceeds remaining budget balance.
 - **6a2.** System triggers **UC-07: Flag Overdraft Warning**.
 - **6a3.** System displays error: *"Transaction failed: Expense amount exceeds remaining grant balance."*
 - **6a4.** System prompts user to modify amount or request Research Dean override.
 - **6a5.** If user cancels, expense is discarded, no changes are committed to ledger, and use case ends.
- **4a. Missing Required Data or Invalid Attachment**
 - **4a1.** System flags missing/invalid input fields.
 - **4a2.** System displays inline error indicators.
 - **4a3.** Faculty Researcher corrects inputs and re-submits.
 - **4a4.** Process resumes at Step 5.